

matchgroup_test_routed

Design Report

Rev 1 | 2026-06-16 | jlcpcb-tier1

kicad-tools 0.13.0

Board Summary

Property	Value
Layers	4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu)
Footprints	8 (5 SMD, 3 THT, 0 other)
Nets	34
Traces	1269 segments
Vias	243
Board Size	110.0 x 95.0 mm

Design Overview

Theory of Operation

Match-Group Test Board

N-trace + group-of-pairs match-group regression testbench

Epic #2661 Phase 3L (issue #2724)

Power Architecture

Power Rails: PWR_FLAG

Assembly Notes

5 fine-pitch components

- **Fine-pitch components:** 5 (U2, U1, U5, U4, U3)

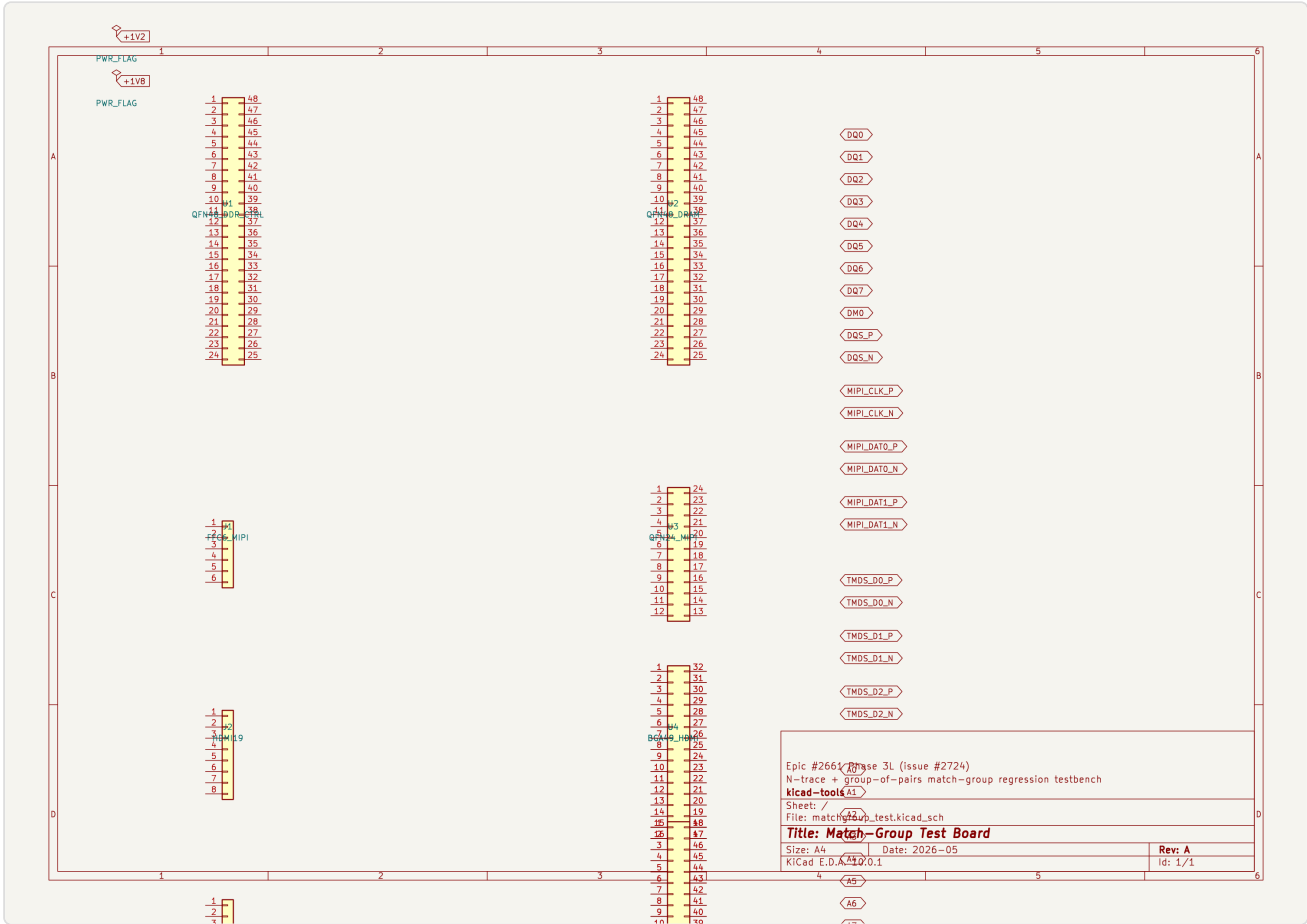
ERC Status

Metric	Count
Errors	0
Warnings	0

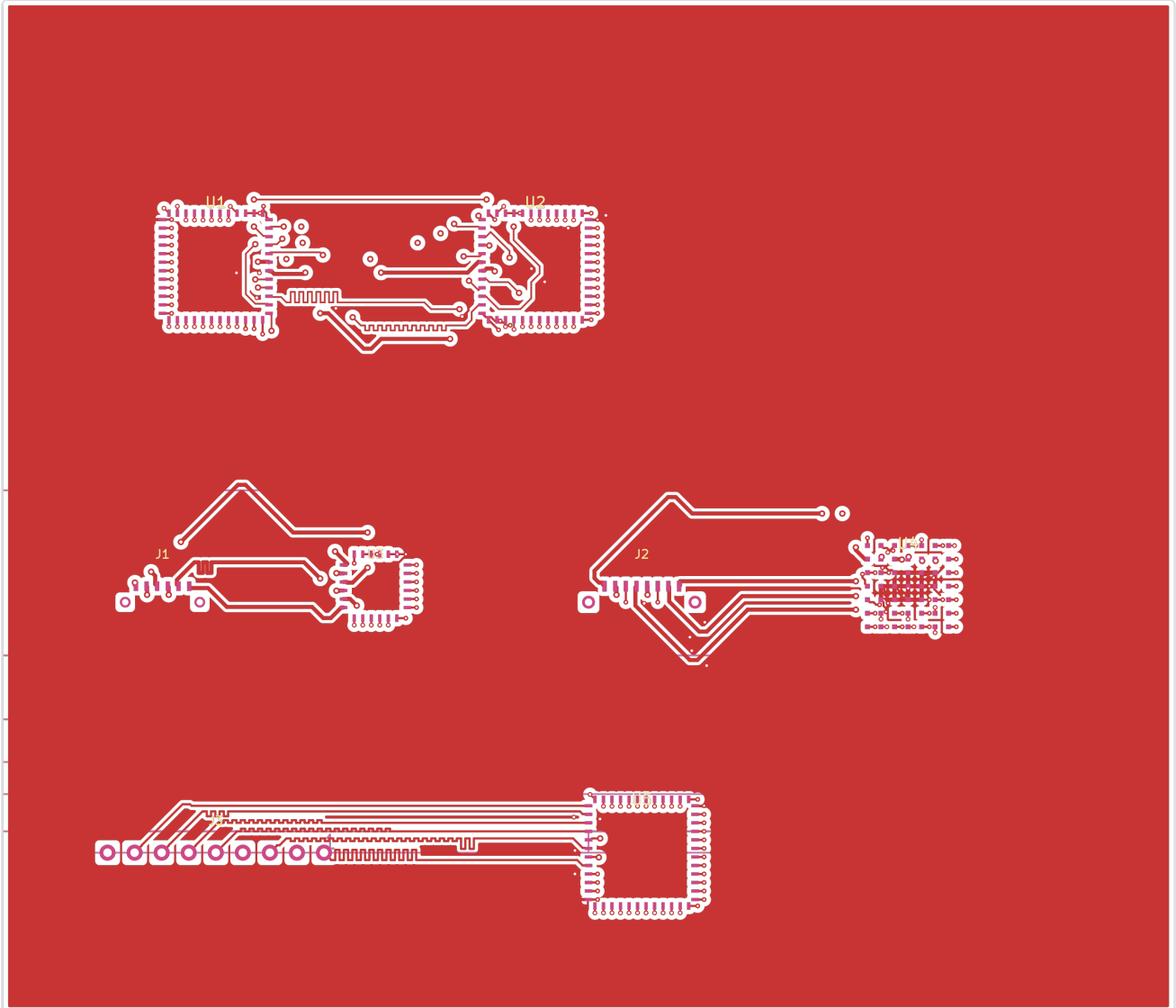
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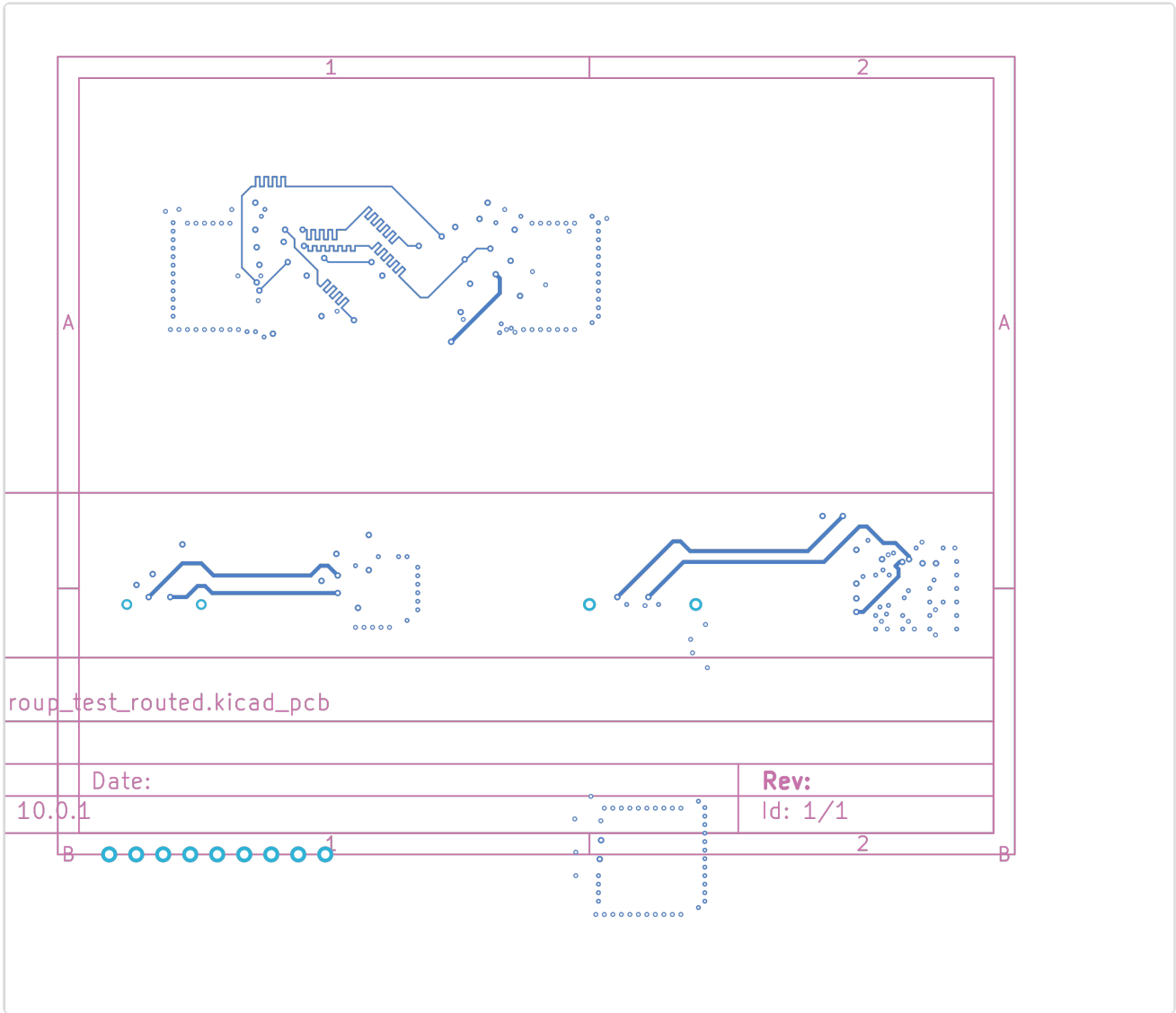
Schematic Overview

Schematic: matchgroup_test

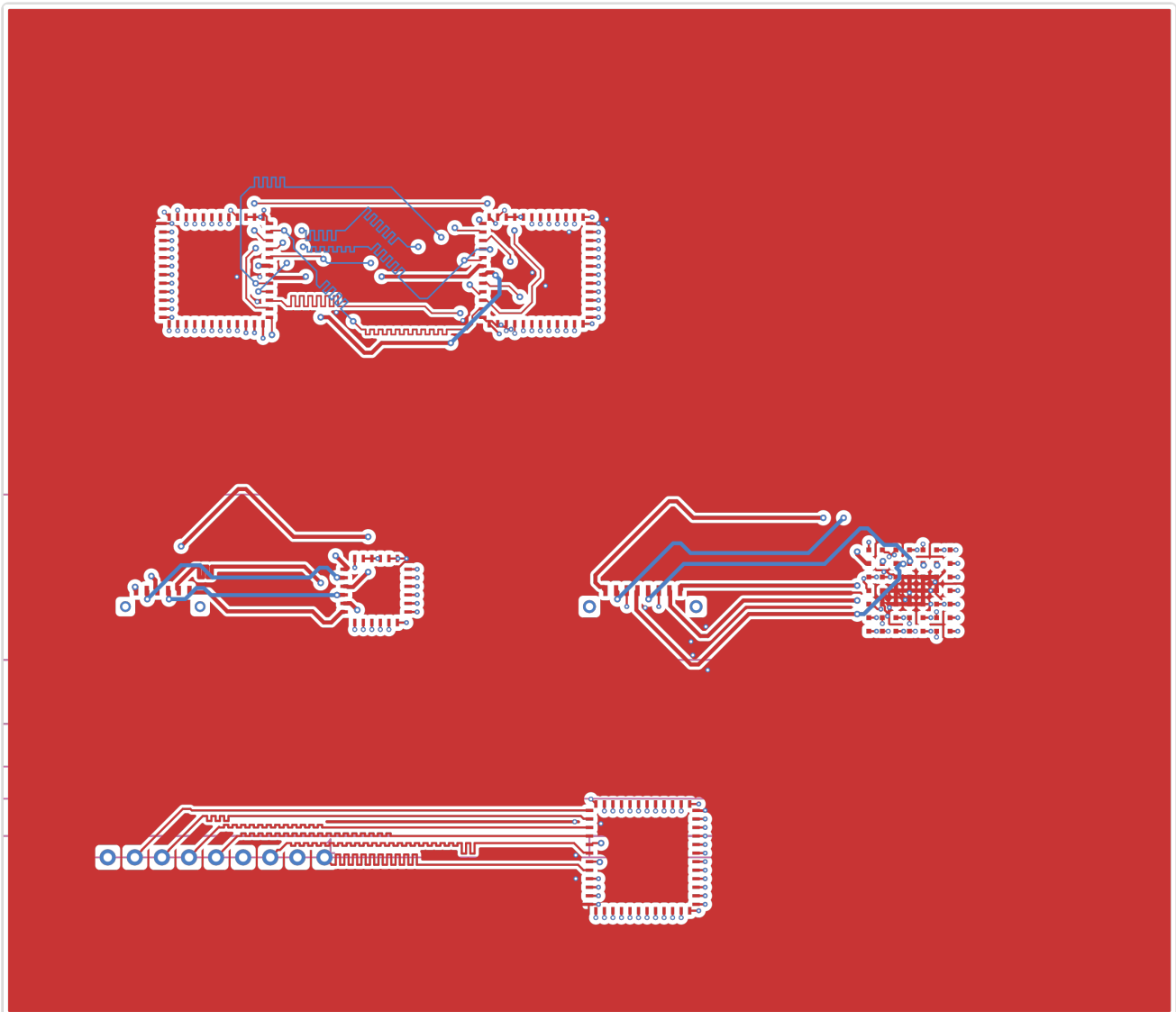


PCB Layout

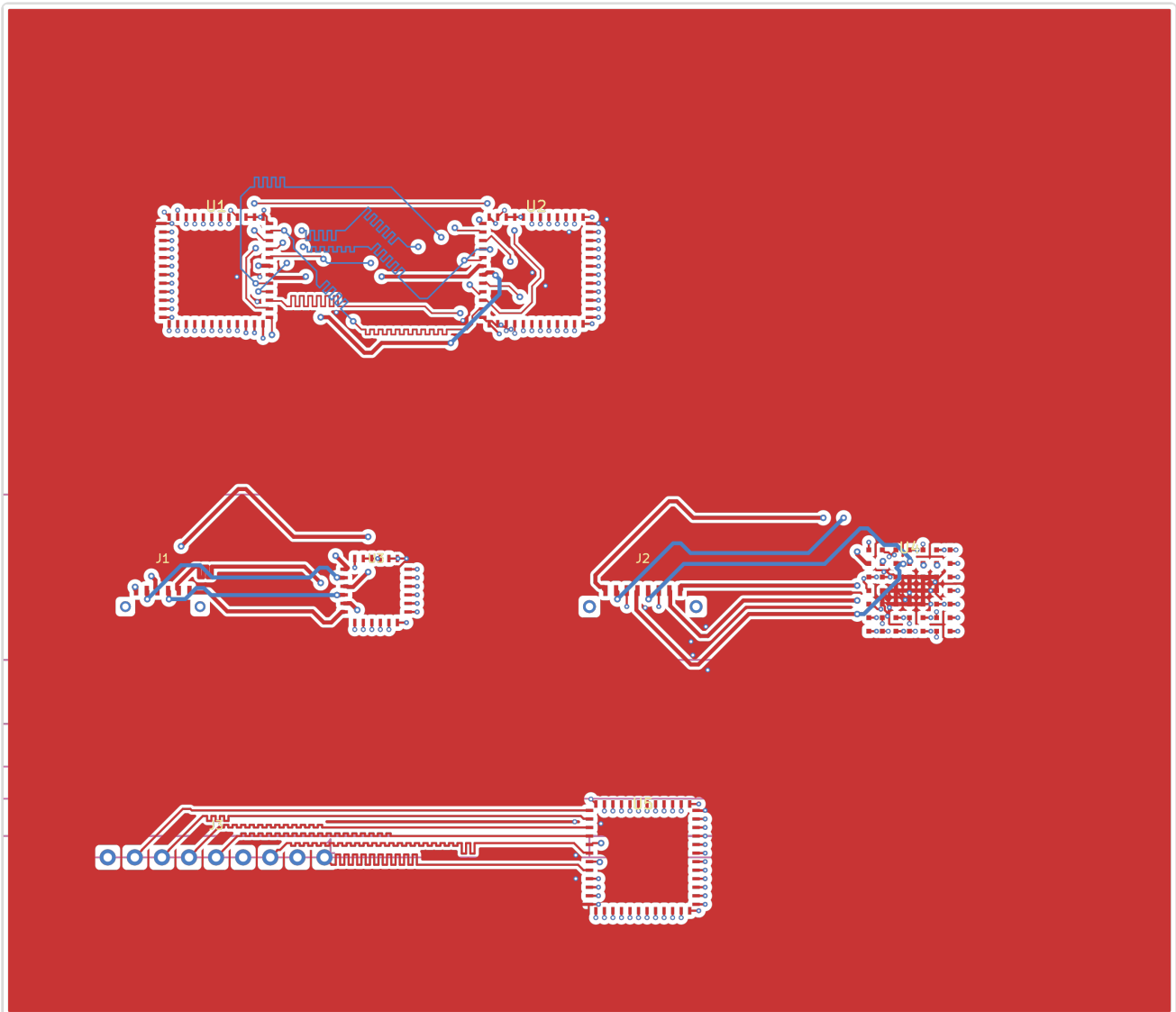




Copper

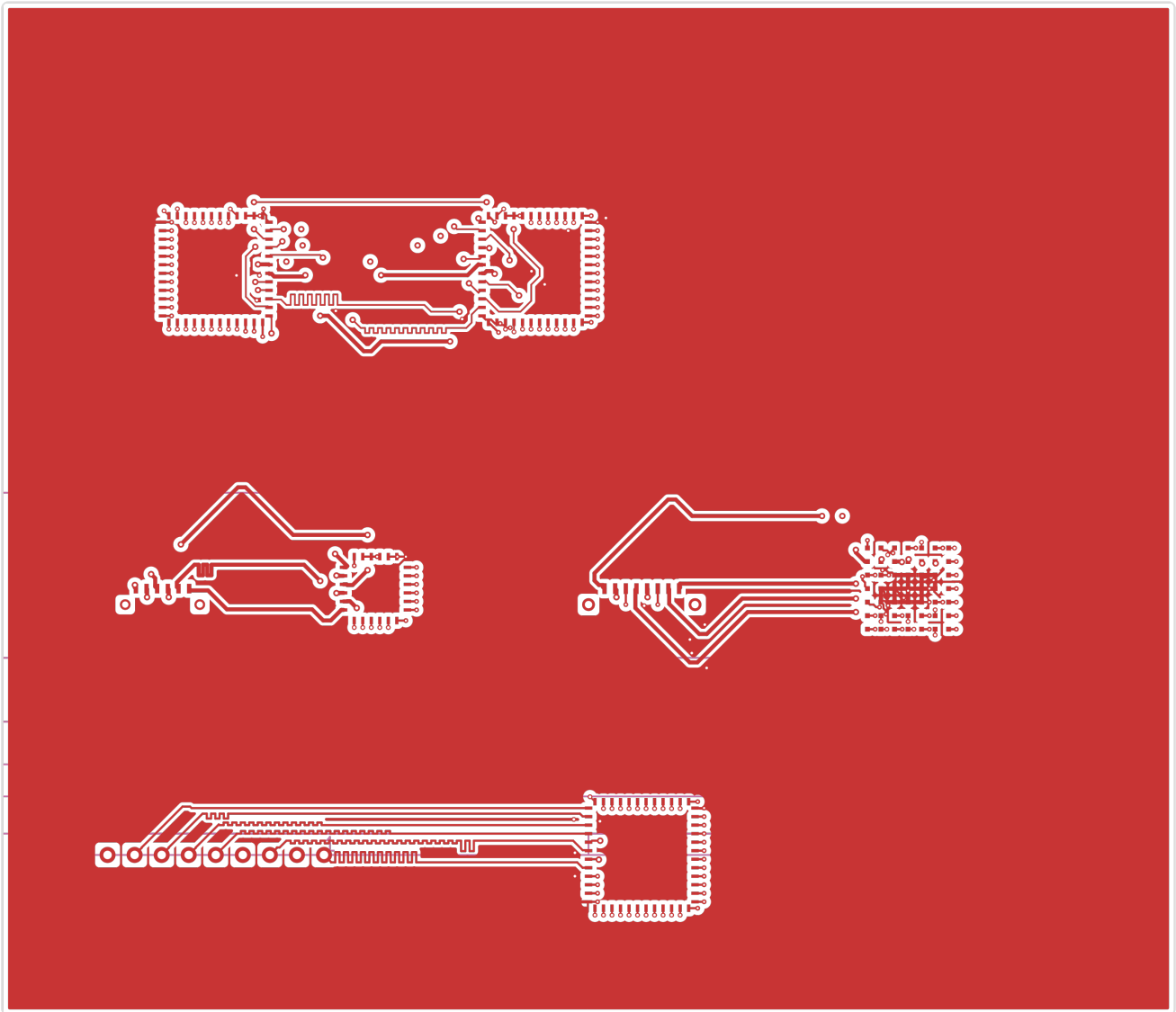


Assembly

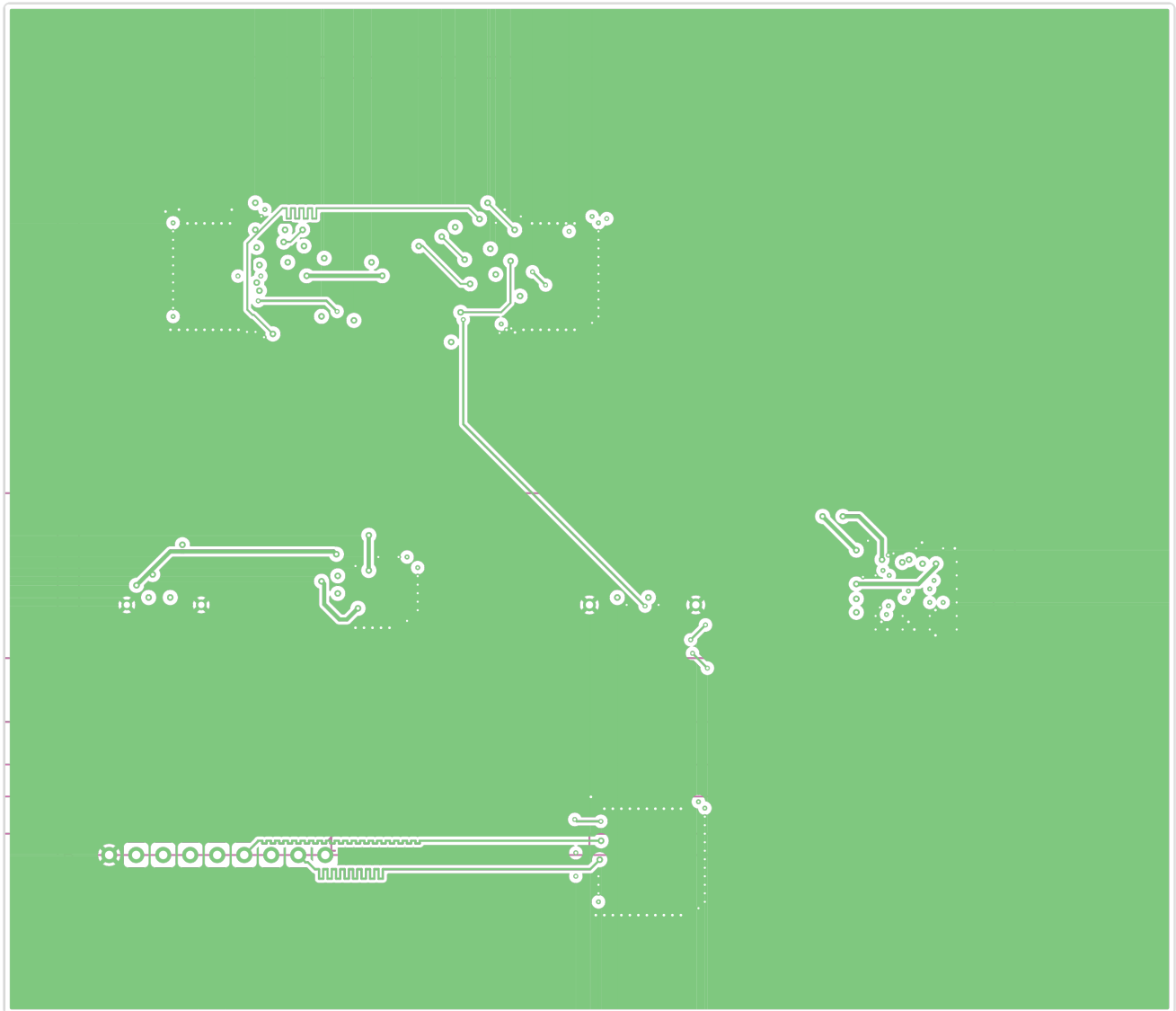


Copper Layers

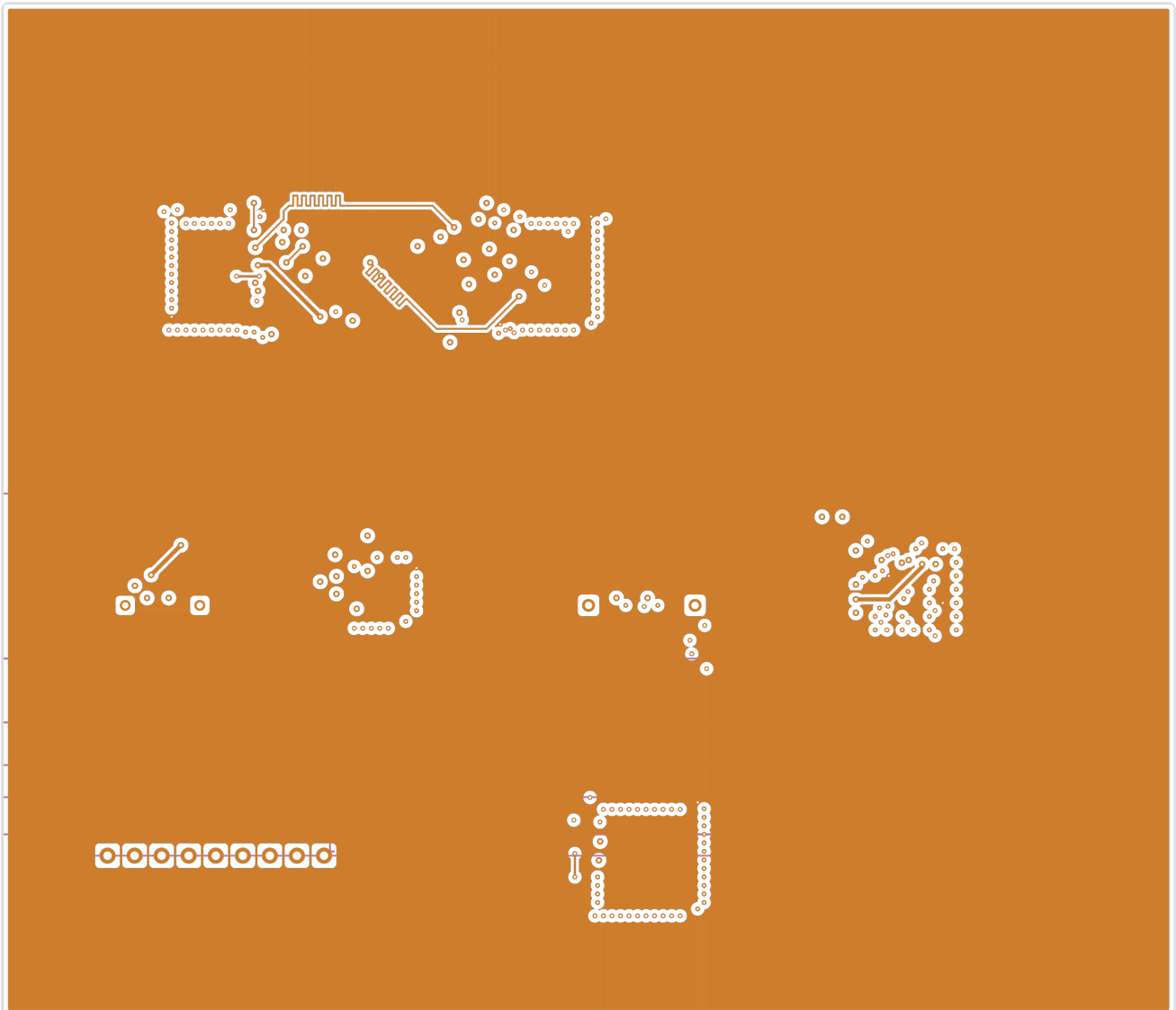
F.Cu



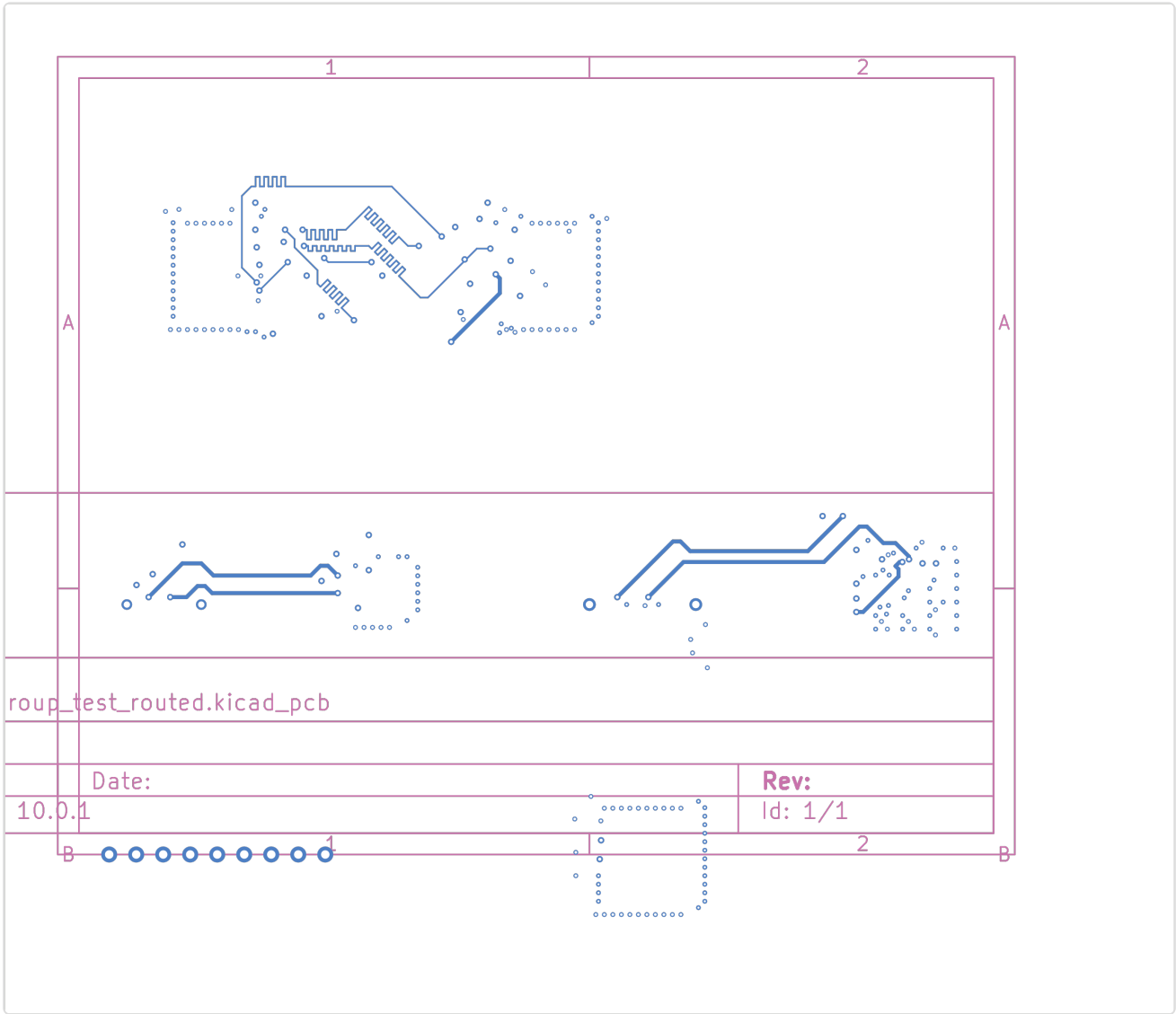
In1.Cu



In2.Cu



B.Cu



Bill of Materials

Value	Package	Qty	References
ADDR_HDR	PinHeader_1x09_P2.54mm_Vertical	1	J3
FFC6_MIPI	FFC_6P_1.0mm	1	J1
HDMI19	HDMI_A_Receptacle	1	J2
BGA49_HDMI	BGA-49_5.0x5.0mm_La yout7x7_P0.5mm	1	U4
QFN24_MIPI	QFN-24-1EP_4x4mm_P 0.5mm	1	U3
QFN48_DDR_CTRL	QFN-48-1EP_7x7mm_P 0.5mm	1	U1
QFN48_DRAM	QFN-48-1EP_7x7mm_P 0.5mm	1	U2
QFP48_SRAM	LQFP-48_7x7mm_P0.5 mm	1	U5

DRC Status

Metric	Count
Errors	125
Warnings	5
Blocking	125

Status: FAIL

Violations by Type

Violation Type	Count
kicad-cli:clearance	26
clearance_segment_via	23
kicad-cli:hole_clearance	18
kicad-cli:shorting_items	13
clearance_pad_via	12
clearance_pad_segment	10
kicad-cli:solder_mask_bridge	8
clearance_segment_segment	7
silkscreen_text_height	4
kicad-cli:zones_intersect	3
clearance_via_via	2
kicad-cli:tracks_crossing	2
connectivity	1
silkscreen_over_pad	1
kicad-cli:starved_thermal	1

Manufacturing Readiness

Verdict: NOT_READY

Action Items

- **[CRITICAL]** Fix 125 blocking DRC violations (clearance_segment_via (23), clearance_pad_via (12), clearance_pad_segment (10); kicad-cli: clearance (26), hole_clearance (18), shorting_items (13))
- **[OPTIONAL]** Verify zone fill in KiCad for 3 zone-connected nets
- **[OPTIONAL]** Review 5 DRC warnings

Routing Status

Metric	Value
Signal Net Completion	100.0% (31/31)
Overall Completion	97.1%
Complete Nets	33 / 34
Zone-Connected Nets	3
Incomplete Nets	1
Unconnected Pads	6

Zone-Connected Nets

- +1V2
- +1V8
- GND

Incomplete Nets

- GND

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~4.09 USD
Components (estimated)	~2.8 USD
Assembly (estimated)	~2.05 USD
Total (estimated)	~8.94 USD
Batch Quantity	5
Batch Total (estimated)	~44.72 USD